

SIMATS ENGINEERING ASSESSMENT TOOL 1

**COURSE NAME: CSA07 COURSE CODE: Computer
Networks**

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable
data transmission across

networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A SHORT ANSWER TEST

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Code of Conduct:

I **T.GANESH(192425246)** (Name / Reg No) certify that this
submission is my original work

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assessment.

I understand that any violation of academic integrity rules will
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Question 1

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport

Layer divides the file into 1500-byte segments, while the Data Link Layer

encapsulates each segment into frames by adding 40 bytes of header and

trailer. Calculate the total number of segments and frames transmitted, the total

transmission overhead introduced by the Data Link Layer, and estimate the total

transmission time (ignoring propagation delay). Explain how the Transport and

Data Link layers ensure reliable communication.

Problem Statement

A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a **bandwidth of 50 Mbps**.

- Transport Layer divides the file into **1500-byte segments**.
- Data Link Layer adds **40 bytes** (header + trailer) to each segment.

- Calculate:
 1. Total number of segments and frames transmitted.
 2. Total transmission overhead introduced by the Data Link Layer.
 3. Total transmission time (ignore propagation delay).
 4. Explain how the Transport Layer and Data Link Layer ensure reliable communication.

Solution

Step 1: Given Data

Parameter	Value
File Size	150 MB
Segment Size	1500 bytes
Data Link Overhead	40 bytes/frame
Bandwidth	50 Mbps

Step 2: Convert File Size into Bytes

We know,

$$1 \text{ MB} = 1024 \times 1024 = 1,048,576 \text{ bytes}$$

Therefore,

$$150 \times 1,048,576 = 157,286,400 \text{ bytes}$$

Total File Size = 157,286,400 bytes

Step 3: Calculate Number of Segments

Formula:

$$\text{Number of Segments} = \frac{\text{Total File Size}}{\text{Segment Size}}$$

Substitute values:

$$\begin{aligned} &= \frac{157,286,400}{1500} \\ &= 104,857.6 \end{aligned}$$

Since partial segments are not possible,

$$\boxed{\text{Total Segments} = 104,858}$$

Step 4: Calculate Number of Frames

Each segment is encapsulated into **one frame**.

Therefore,

$$\boxed{\text{Total Frames} = 104,858}$$

Step 5: Calculate Total Data Link Layer Overhead

Each frame adds

40 bytes

Formula:

$$\begin{aligned}\text{Overhead} &= 104,858 \times 40 \\ &= 4,194,320 \text{ bytes}\end{aligned}$$

Convert into MB:

$$\frac{4,194,320}{1,048,576} = 4 \text{ MB}$$

$\text{Total Overhead} = 4,194,320 \text{ bytes } (\approx 4 \text{ MB})$

Step 6: Calculate Total Data Transmitted

Actual file

$$157,286,400$$

•

Overhead

$$4,194,320$$

=

$$161,480,720 \text{ bytes}$$

Convert to bits

$$161,480,720 \times 8 = 1,291,845,760 \text{ bits}$$

Step 7: Calculate Transmission Time

Bandwidth

$$50 \text{ Mbps} = 50,000,000 \text{ bits/sec}$$

Formula

$$\begin{aligned}\text{Transmission Time} &= \frac{\text{Total Bits}}{\text{Bandwidth}} \\ &= \frac{1,291,845,760}{50,000,000} \\ &= 25.84 \text{ seconds}\end{aligned}$$

Transmission Time $\approx$ 25.84 seconds

Final Answers

Calculation	Answer
Number of Segments	104,858
Number of Frames	104,858
Data Link Layer Overhead	4,194,320 bytes ($\approx$ 4 MB)
Total Transmission Time	25.84 seconds

Concept Explanation

Role of the Transport Layer

The **Transport Layer (Layer 4)** is responsible for reliable end-to-end communication.

Its functions include:

- Dividing large files into smaller **segments**.
- Assigning sequence numbers.

- Detecting lost segments.
- Retransmitting missing data.
- Performing flow control.
- Ensuring error-free delivery.

Protocols used:

- TCP
- UDP

For reliable file transfer, **TCP** is commonly used.

Role of the Data Link Layer

The **Data Link Layer (Layer 2)** is responsible for node-to-node communication.

Its functions include:

- Encapsulating each segment into a frame.
- Adding **header and trailer**.
- Detecting transmission errors using **CRC (Cyclic Redundancy Check)**.
- Controlling access to the communication medium.
- Retransmitting damaged frames if necessary.

Application

Machine Learning, Cloud Computing, Banking Systems, Healthcare Networks, and Video Streaming services rely on

the Transport and Data Link layers to ensure reliable communication and accurate data delivery.

Conclusion

The 150 MB file is divided into **104,858 segments**, and the same number of frames are transmitted. The Data Link Layer introduces approximately **4 MB of overhead**, resulting in a total transmission time of **25.84 seconds** over a **50 Mbps** network. The **Transport Layer** ensures reliable end-to-end delivery using segmentation, acknowledgments, and retransmissions, while the **Data Link Layer** provides error detection and reliable frame transmission between adjacent devices.

Question 2

Problem Statement

A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate:

1. Number of transmission windows required.
2. Total number of retransmissions if **one frame in every window is lost**.
3. Explain how **flow control** improves communication efficiency.

Solution

Step 1: Given Data

Parameter	Value
Sliding Window Size	6 frames
Data per Frame	1024 bytes
Total Frames	48 frames
Lost Frames	1 frame per window

Step 2: Calculate Number of Transmission Windows

Formula

$$\text{Number of Windows} = \frac{\text{Total Frames}}{\text{Window Size}}$$

Calculation

$$\begin{aligned} &= \frac{48}{6} \\ &= 8 \end{aligned}$$

Answer

Number of Transmission Windows = 8

Step 3: Calculate Total Data Transmitted

Each frame contains **1024 bytes**.

Formula

$$\text{Total Data} = 48 \times 1024$$

Calculation

$$= 49,152 \text{ bytes}$$

Answer

Total User Data = 49,152 bytes

Step 4: Calculate Retransmissions

There are **8 transmission windows**.

One frame is lost in **every window**.

Formula

$$\text{Retransmissions} = \text{Number of Windows} \times 1$$

Calculation

$$\begin{aligned} &= 8 \times 1 \\ &= 8 \end{aligned}$$

Answer

Total Retransmissions = 8 frames

Step 5: Total Frames Sent Including Retransmissions

Original Frames

$$48$$

Retransmitted Frames

8

Total Frames

$$\begin{aligned} &48 + 8 \\ &= 56 \end{aligned}$$

Answer

Total Frames Transmitted = 56

Final Answers

Calculation	Answer
Sliding Window Size	6 frames
Total Frames	48
Number of Windows	8
Retransmissions	8 frames
Total Frames Sent	56 frames
Total User Data	49,152 bytes

Concept Explanation

What is Sliding Window?

The **Sliding Window Protocol** is a **Transport Layer** mechanism that allows the sender to transmit multiple frames before receiving acknowledgements. This increases network utilization and reduces waiting time.

Working Process

1. Sender transmits **6 frames** (Window Size = 6).
2. Receiver acknowledges the received frames.
3. The transmission window "slides" forward.
4. If a frame is lost, only that frame is retransmitted (depending on the protocol, such as Selective Repeat; Go-Back-N may retransmit additional frames).

Flow Control

Flow control prevents the sender from transmitting data faster than the receiver can process it.

Importance of Flow Control

- Prevents receiver buffer overflow.
- Reduces packet loss.
- Improves communication reliability.
- Maintains smooth data transfer.
- Increases overall network efficiency.

Example: If a sender is much faster than the receiver, flow control temporarily limits the sending rate until the receiver is ready.

Real-World Applications

Sliding Window and Flow Control are used in:

- File Transfer Protocol (FTP)
- Email communication
- Web browsing (HTTP over TCP)
- Video streaming
- Cloud storage
- Online banking
- Database synchronization

Conclusion

The sender requires **8 transmission windows** to send **48 frames**. Since **one frame is lost in each window**, **8 retransmissions** are needed, making the total number of transmitted frames **56**. The **Sliding Window Protocol** improves network efficiency by allowing multiple frames to be sent before acknowledgements are received, while **flow control** ensures that the receiver is not overwhelmed, leading to reliable and efficient communication.

Question 3

Problem Statement

A **12,000-byte IP packet** is transmitted through a router where the **Maximum Transmission Unit (MTU)** is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate:

1. The total number of fragments created.
2. The total IP header overhead introduced.
3. Explain how the **Network Layer** ensures successful delivery of fragmented packets to the destination.

Solution

Step 1: Given Data

Parameter	Value
Original IP Packet Size	12,000 bytes
MTU	1500 bytes
IP Header Size	20 bytes

Step 2: Calculate Maximum Data per Fragment

Each fragment must include a **20-byte IP header**.

Therefore, the maximum data that can be carried in one fragment is:

Formula

$$\text{Maximum Data per Fragment} = \text{MTU} - \text{IP Header}$$

Calculation

$$\begin{aligned} &= 1500 - 20 \\ &= 1480 \text{ bytes} \end{aligned}$$

Answer

$$\text{Maximum Data per Fragment} = 1480 \text{ bytes}$$

Step 3: Calculate Number of Fragments

The original packet contains **12,000 bytes of data**.

Formula

$$\text{Number of Fragments} = \left\lceil \frac{\text{Packet Size}}{\text{Data per Fragment}} \right\rceil$$

Calculation

$$= \frac{12000}{1480} = 8.108$$

Since a partial fragment is not possible,

$$\text{Number of Fragments} = 9$$

Step 4: Fragment Distribution

Fragment	Data Size (Bytes)	Header (Bytes)	Total Fragment Size
1	1480	20	1500
2	1480	20	1500
3	1480	20	1500

Fragment	Data Size (Bytes)	Header (Bytes)	Total Fragment Size
4	1480	20	1500
5	1480	20	1500
6	1480	20	1500
7	1480	20	1500
8	1480	20	1500
9	160	20	180

Verification:

$$1480 \times 8 + 160 = 12,000 \text{ bytes}$$

✓ Correct.

Step 5: Calculate Total Header Overhead

Each fragment requires a **20-byte IP header**.

Formula

$$\begin{aligned} \text{Total Header Overhead} \\ &= \text{Number of Fragments} \times \text{Header Size} \end{aligned}$$

Calculation

$$\begin{aligned} &= 9 \times 20 \\ &= 180 \text{ bytes} \end{aligned}$$

Answer

$$\boxed{\text{Total Header Overhead} = 180 \text{ bytes}}$$

Step 6: Total Data Transmitted

Original Data

12000 bytes

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Header Overhead

180 bytes

=

12180 bytes

Answer

$$\boxed{\text{Total Data Transmitted} = 12,180 \text{ bytes}}$$

Final Answers

Calculation

Answer

Maximum Data per Fragment **1480 bytes**

Number of Fragments **9**

Total Header Overhead **180 bytes**

Calculation

Answer

Total Data Transmitted

12,180 bytes

Concept Explanation

What is Fragmentation?

Fragmentation is the process of dividing a large IP packet into smaller packets (fragments) when it is larger than the network's **Maximum Transmission Unit (MTU)**.

Each fragment:

- Has its own IP header.
- Travels independently through the network.
- Is reassembled at the destination.

Role of the Network Layer (Layer 3)

The **Network Layer** is responsible for:

- Assigning logical IP addresses.
- Determining the best routing path.
- Fragmenting packets when necessary.
- Delivering packets to the destination.
- Reassembling fragments at the destination.

How the Network Layer Ensures Successful Delivery

1. **Fragmentation:** Divides large packets into smaller fragments based on the MTU.

2. **Routing:** Sends each fragment through the best available path.
3. **Identification:** Adds an Identification field so all fragments of the same packet can be recognized.
4. **Fragment Offset:** Indicates the position of each fragment in the original packet.
5. **More Fragments (MF) Flag:** Indicates whether more fragments follow.
6. **Reassembly:** The destination host uses the Identification, Fragment Offset, and MF flag to correctly reconstruct the original packet.

Real-World Applications

Fragmentation is commonly used in:

- Internet communication
- Cloud computing
- Video streaming
- File transfer (FTP)
- VPNs
- Mobile networks (4G/5G)
- IoT devices

Conclusion

A 12,000-byte IP packet transmitted over a network with an **MTU of 1500 bytes** is divided into **9 fragments**. Each

fragment contains a **20-byte IP header**, resulting in a total header overhead of **180 bytes**. Therefore, the total amount of data transmitted is **12,180 bytes**. The **Network Layer** ensures reliable delivery through fragmentation, routing, and reassembly at the destination.

Question 4

Problem Statement

A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate:

1. Number of transmission windows required.
2. Total number of retransmissions if **one frame in every window is lost**.
3. Explain how **flow control** improves communication efficiency.

Solution

Step 1: Given Data

Parameter	Value
Sliding Window Size	6 frames
User Data per Frame	1024 bytes
Total Frames	48 frames
Lost Frames	1 frame per window

Step 2: Calculate Number of Transmission Windows

Formula

$$\text{Number of Windows} = \frac{\text{Total Frames}}{\text{Window Size}}$$

Calculation

$$\begin{aligned} &= \frac{48}{6} \\ &= 8 \end{aligned}$$

Answer

Number of Transmission Windows = 8

Step 3: Calculate Total User Data

Formula

$$\text{Total User Data} = \text{Total Frames} \times \text{Data per Frame}$$

Calculation

$$\begin{aligned} &= 48 \times 1024 \\ &= 49,152 \text{ bytes} \end{aligned}$$

Answer

Total User Data = 49,152 bytes

Step 4: Calculate Total Retransmissions

Since one frame is lost in every window:

Formula

$$\text{Retransmissions} = \text{Number of Windows} \times 1$$

Calculation

$$= 8 \times 1$$

$$= 8$$

Answer

$\text{Total Retransmissions} = 8 \text{ frames}$

Step 5: Calculate Total Frames Sent

Original Frames:

$$48$$

Retransmitted Frames:

$$8$$

Total Frames Sent:

$$48 + 8 = 56$$

Answer

$$\boxed{\text{Total Frames Sent} = 56}$$

Final Answers

Calculation	Answer
Total Frames	48
Window Size	6
Number of Windows	8
Retransmissions	8 frames
Total Frames Sent	56 frames
Total User Data	49,152 bytes

Concept Explanation

Sliding Window Protocol

The **Sliding Window Protocol** is a **Transport Layer** mechanism that allows the sender to transmit multiple frames before waiting for acknowledgements.

Working Process

1. The sender sends **6 frames**.
2. The receiver acknowledges the received frames.
3. The transmission window moves (slides) forward.
4. If one frame is lost, it is retransmitted.

This process continues until all frames are successfully delivered.

Flow Control

Flow Control regulates the rate of data transmission so that the sender does not send data faster than the receiver can process it.

Importance of Flow Control

- Prevents receiver buffer overflow.
- Reduces packet loss.
- Improves communication efficiency.
- Ensures reliable data transfer.
- Maintains smooth network performance.

Real-World Applications

The Sliding Window Protocol and Flow Control are used in:

- File Transfer Protocol (FTP)
- Web Browsing (HTTP over TCP)
- Email Communication
- Video Streaming
- Cloud Storage
- Online Banking
- Database Synchronization

Conclusion

A total of **48 frames** are transmitted using a **sliding window size of 6**, requiring **8 transmission windows**. Since **one frame is lost in each window**, **8 retransmissions** are required, making the total number of transmitted frames **56**. The **Sliding Window Protocol** improves network utilization by allowing multiple frames to be sent before acknowledgements are received, while **Flow Control** ensures reliable and efficient communication by preventing the sender from overwhelming the receiver.

Question 5

Problem Statement

A multimedia application transfers a **600 MB** video file. The **Session Layer** inserts a **synchronization checkpoint after every 60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred.

Calculate:

1. The number of checkpoints created before the failure.
2. The amount of data that must be retransmitted after recovery.
3. Explain the importance of synchronization in the Session Layer.

Solution

Step 1: Given Data

Parameter	Value
Total Video File Size	600 MB
Checkpoint Interval	60 MB
Data Transferred Before Failure	390 MB

Step 2: Calculate Number of Checkpoints

A checkpoint is created **after every 60 MB**.

Formula

$$\text{Number of Checkpoints} = \left\lfloor \frac{\text{Data Transferred}}{\text{Checkpoint Interval}} \right\rfloor$$

Calculation

$$\begin{aligned} &= \frac{390}{60} \\ &= 6.5 \end{aligned}$$

Only **completed checkpoints** are counted.

Therefore,

$$\boxed{\text{Number of Checkpoints} = 6}$$

The checkpoints are created at:

- 60 MB
- 120 MB
- 180 MB

- 240 MB
- 300 MB
- 360 MB

Step 3: Calculate Data to be Retransmitted

Failure occurs at **390 MB**.

The last completed checkpoint is at **360 MB**.

Therefore,

Formula

$$\text{Retransmitted Data} = 390 - 360$$

Calculation

$$= 30 \text{ MB}$$

Answer

$\text{Data to be Retransmitted} = 30 \text{ MB}$

Step 4: Remaining Data After Recovery

Remaining file size:

$$600 - 390 = 210 \text{ MB}$$

After retransmitting the lost **30 MB**, the system continues with the remaining **210 MB**.

Final Answers

Calculation	Answer
Total File Size	600 MB
Checkpoint Interval	60 MB
Failure Occurs At	390 MB
Checkpoints Created	6
Last Checkpoint	360 MB
Data Retransmitted	30 MB
Remaining Data to Transfer	210 MB

Concept Explanation

What is the Session Layer?

The **Session Layer (Layer 5)** of the OSI Model is responsible for establishing, managing, and terminating communication sessions between two devices.

Its main functions include:

- Session establishment
- Session maintenance
- Session termination
- Synchronization
- Recovery from communication failures

What is Synchronization?

Synchronization involves placing **checkpoints** in a data stream during transmission.

These checkpoints act as recovery points.

If communication fails, transmission resumes from the **last checkpoint** instead of starting from the beginning.

Importance of Synchronization

Synchronization provides several benefits:

- Reduces retransmission time.
- Saves network bandwidth.
- Improves reliability.
- Enables faster recovery after failures.
- Prevents unnecessary retransmission of already received data.

Real-World Applications

Session Layer synchronization is used in:

- Video streaming
- Large file downloads
- Cloud storage services
- Online backup systems
- Database replication
- Video conferencing

- Banking transaction systems

Illustration

600 MB Video File

0 MB

|

|——60 MB ✓ Checkpoint 1

|——120 MB ✓ Checkpoint 2

|——180 MB ✓ Checkpoint 3

|——240 MB ✓ Checkpoint 4

|——300 MB ✓ Checkpoint 5

|——360 MB ✓ Checkpoint 6

|

|——390 MB ✗ Failure Occurs

|

|—— Retransmit only 30 MB

Conclusion

A **600 MB** video file is transmitted with synchronization checkpoints every **60 MB**. Before the failure at **390 MB**, **6 checkpoints** are successfully created. Since the last checkpoint is at **360 MB**, only **30 MB** of data needs to be

retransmitted. The **Session Layer** improves communication by enabling synchronization and recovery, reducing transmission time, saving bandwidth, and ensuring reliable data transfer.

Question 6

Problem Statement

Before transmission, the **Presentation Layer** compresses a **900 MB** file by **40%**. The compressed data is then encrypted, adding **5% overhead** to the compressed size.

Calculate:

1. The compressed file size.
2. The encrypted file size.
3. The total percentage reduction compared to the original file.
4. Explain how the **Presentation Layer** improves transmission efficiency and security.

Solution

Step 1: Given Data

Parameter	Value
Original File Size	900 MB
Compression	40%
Encryption Overhead	5% of compressed size

Step 2: Calculate Compressed File Size

Formula

$$\begin{aligned}\text{Compressed Size} \\ &= \text{Original Size} \times (1 - \text{Compression Percentage})\end{aligned}$$

Calculation

$$\begin{aligned}&= 900 \times (1 - 0.40) \\ &= 900 \times 0.60 \\ &= 540 \text{ MB}\end{aligned}$$

Answer

Compressed File Size = 540 MB

Step 3: Calculate Encryption Overhead

Encryption adds **5%** overhead to the compressed file.

Formula

$$\text{Encryption Overhead} = \text{Compressed Size} \times 5\%$$

Calculation

$$\begin{aligned}&= 540 \times 0.05 \\ &= 27 \text{ MB}\end{aligned}$$

Answer

Encryption Overhead = 27 MB

Step 4: Calculate Encrypted File Size

Formula

$$\begin{aligned}\text{Encrypted File Size} \\ &= \text{Compressed Size} + \text{Encryption Overhead}\end{aligned}$$

Calculation

$$\begin{aligned}&= 540 + 27 \\ &= 567 \text{ MB}\end{aligned}$$

Answer

$\text{Encrypted File Size} = 567 \text{ MB}$

Step 5: Calculate Total Percentage Reduction

Original File Size

$$= 900 \text{ MB}$$

Final File Size

$$= 567 \text{ MB}$$

Reduction in Size

$$900 - 567 = 333 \text{ MB}$$

Formula

$$\text{Percentage Reduction} = \frac{\text{Reduction}}{\text{Original Size}} \times 100$$

Calculation

$$\begin{aligned} &= \frac{333}{900} \times 100 \\ &= 37\% \end{aligned}$$

Answer

Total Percentage Reduction = 37%

Final Answers

Calculation	Answer
Original File Size	900 MB
Compressed File Size	540 MB
Encryption Overhead	27 MB
Encrypted File Size	567 MB
Total Reduction	333 MB
Percentage Reduction	37%

Concept Explanation

What is the Presentation Layer?

The **Presentation Layer (Layer 6)** of the OSI Model is responsible for preparing data before it is transmitted over the network.

Its major functions include:

- Data Compression
- Data Encryption
- Data Decryption
- Data Translation
- Data Formatting

How the Presentation Layer Improves Transmission Efficiency

1. Data Compression

- Reduces the size of data before transmission.
- Uses less network bandwidth.
- Speeds up data transfer.
- Reduces storage requirements.

Example: Compressing a 900 MB file to 540 MB.

2. Data Encryption

- Converts readable data into an unreadable format (cipher text).
- Protects data from unauthorized access.

- Ensures confidentiality during transmission.

Example: HTTPS uses encryption to protect online banking and shopping transactions.

3. Data Translation

The Presentation Layer converts data into a standard format so that devices using different operating systems can communicate correctly.

Real-World Applications

The Presentation Layer is used in:

- HTTPS (Secure Web Browsing)
- Online Banking
- E-commerce Websites
- Cloud Storage
- Email Encryption
- Video Compression
- File Compression (ZIP, RAR)

Illustration

Original File

900 MB



Compression (40%)

|



540 MB

|

Encryption (+5%)

|



567 MB

|

Secure Transmission

Conclusion

The **Presentation Layer** compresses the original **900 MB** file to **540 MB**, reducing the amount of data that needs to be transmitted. Encryption then adds **27 MB** of overhead, resulting in a final encrypted file size of **567 MB**. Compared to the original file, this represents a **37% reduction** in data size. By combining **compression** and **encryption**, the Presentation Layer improves both **transmission efficiency** and **data security**, ensuring faster and safer communication.

Question 7

Problem Statement

An **FTP application** transfers **3 GB** of data over a network with an **effective throughput of 120 Mbps**. The **Application Layer** divides the data into requests of **3 MB** each.

Calculate:

1. Total number of requests generated.
2. Estimate the total transmission time (in seconds).
3. Discuss how the **Application Layer** provides reliable services to users.

Solution

Step 1: Given Data

Parameter	Value
Total Data Size	3 GB
Request Size	3 MB
Throughput	120 Mbps

Step 2: Convert 3 GB into MB

We know,

$$1 \text{ GB} = 1024 \text{ MB}$$

Therefore,

$$3 \times 1024 = 3072 \text{ MB}$$

Answer

$$\boxed{\text{Total Data} = 3072 \text{ MB}}$$

Step 3: Calculate Number of Requests

Formula

$$\text{Number of Requests} = \frac{\text{Total Data}}{\text{Request Size}}$$

Calculation

$$= \frac{3072}{3} = 1024$$

Answer

$$\boxed{\text{Total Requests} = 1024}$$

Step 4: Convert Data into Megabits (Mb)

Since network speed is given in **Mbps**, convert MB to Mb.

We know,

$$1 \text{ MB} = 8 \text{ Mb}$$

Therefore,

$$3072 \times 8 = 24576 \text{ Mb}$$

Answer

$$\boxed{\text{Total Data} = 24,576 \text{ Mb}}$$

Step 5: Calculate Transmission Time

Formula

$$\text{Transmission Time} = \frac{\text{Total Data (Mb)}}{\text{Throughput (Mbps)}}$$

Calculation

$$\begin{aligned} &= \frac{24576}{120} \\ &= 204.8 \text{ seconds} \end{aligned}$$

Answer

$$\boxed{\text{Transmission Time} = 204.8 \text{ seconds}}$$

Final Answers

Calculation	Answer
Total Data	3 GB = 3072 MB
Request Size	3 MB
Number of Requests	1024
Throughput	120 Mbps

Calculation Answer

Transmission Time **204.8 seconds ($\approx$ 3 minutes 25 seconds)**

Concept Explanation

What is the Application Layer?

The **Application Layer (Layer 7)** is the topmost layer of the OSI Model. It provides network services directly to end users and applications.

Main Functions

- File Transfer (FTP)
- Email Services (SMTP)
- Web Browsing (HTTP/HTTPS)
- Remote Login (Telnet/SSH)
- Domain Name Services (DNS)

How the Application Layer Provides Reliable Services

The Application Layer improves reliability by:

1. File Transfer Management

- Divides large files into smaller requests.
- Ensures all file parts are transferred correctly.

2. Error Handling

- Detects incomplete or failed file transfers.
- Allows retransmission of missing data.

3. User Authentication

- Verifies the identity of users before allowing access.
- Prevents unauthorized access to files.

4. Data Integrity

- Ensures that the received file matches the original file.
- Detects corruption during transmission.

5. Session Management

- Establishes and terminates communication sessions.
- Maintains reliable communication between client and server.

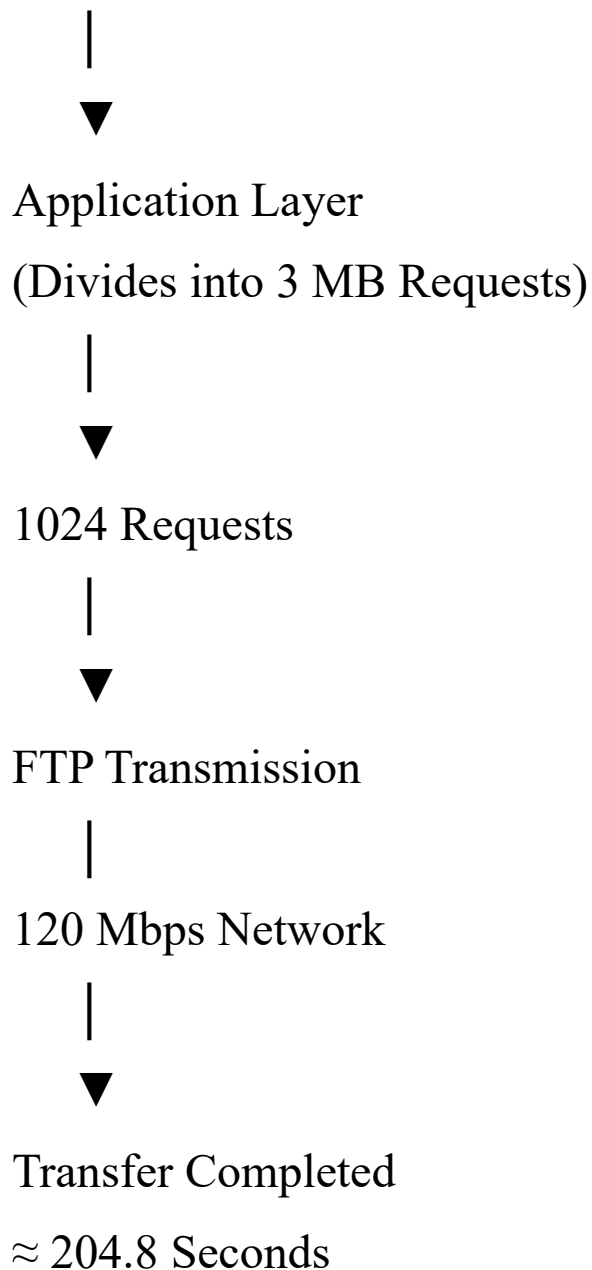
Real-World Applications

The Application Layer is widely used in:

- FTP (File Transfer Protocol)
- Web Browsers (HTTP/HTTPS)
- Email Services
- Cloud Storage (Google Drive, OneDrive)
- Online Banking
- E-commerce Websites
- Video Streaming Platforms

Illustration

3 GB File



Conclusion

The **3 GB** file is divided into **1024 requests**, each containing **3 MB** of data. Over a network with an effective throughput of **120 Mbps**, the total transmission time is **204.8 seconds** (approximately **3 minutes 25 seconds**). The **Application Layer** ensures reliable communication by managing file

transfers, authenticating users, handling errors, maintaining data integrity, and establishing communication sessions.

Question 8

Problem Statement

A packet experiences the following processing delays at different OSI layers:

- **Application Layer = 4 ms**
- **Transport Layer = 3 ms**
- **Network Layer = 5 ms**
- **Data Link Layer = 2 ms**
- **Physical Layer = 1 ms**

Additionally,

- **Propagation Delay = 18 ms**
- **Transmission Delay = 12 ms**

Calculate:

1. The **total end-to-end delay** experienced by the packet.
2. Explain how each OSI layer contributes to the communication process.

Solution

Step 1: Given Data

Parameter	Delay
-----------	-------

Application Layer Delay	4 ms
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Parameter	Delay
Transport Layer Delay	3 ms
Network Layer Delay	5 ms
Data Link Layer Delay	2 ms
Physical Layer Delay	1 ms
Propagation Delay	18 ms
Transmission Delay	12 ms

Step 2: Calculate Total Processing Delay

Processing delay is the sum of delays at all OSI layers.

Formula

$$\begin{aligned}\text{Processing Delay} \\ &= \text{Application} + \text{Transport} + \text{Network} + \text{Data Link} \\ &\quad + \text{Physical}\end{aligned}$$

Calculation

$$\begin{aligned}&= 4 + 3 + 5 + 2 + 1 \\ &= 15 \text{ ms}\end{aligned}$$

Answer

$\text{Processing Delay} = 15 \text{ ms}$

Step 3: Calculate Total End-to-End Delay

Formula

$$\begin{aligned}\text{Total Delay} \\ &= \text{Processing Delay} + \text{Propagation Delay} \\ &+ \text{Transmission Delay}\end{aligned}$$

Calculation

$$\begin{aligned}&= 15 + 18 + 12 \\ &= 45 \text{ ms}\end{aligned}$$

Answer

Total End-to-End Delay = 45 ms

Final Answers

Calculation	Answer
Processing Delay	15 ms
Propagation Delay	18 ms
Transmission Delay	12 ms
Total End-to-End Delay	45 ms

Concept Explanation

How Each OSI Layer Contributes

1. Application Layer (4 ms)

- Provides services to users and applications.
- Examples: Web browsing, Email, FTP.
- Processes user requests before sending data.

2. Transport Layer (3 ms)

- Divides data into segments.
- Provides reliable communication using TCP.
- Performs flow control and error recovery.

3. Network Layer (5 ms)

- Determines the best route for packets.
- Uses logical addressing (IP).
- Performs routing and packet forwarding.

4. Data Link Layer (2 ms)

- Converts packets into frames.
- Performs error detection using CRC.
- Controls communication between adjacent devices.

5. Physical Layer (1 ms)

- Converts data into electrical, optical, or radio signals.
- Transmits bits through cables or wireless media.

Additional Delays

Propagation Delay (18 ms)

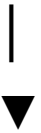
- Time taken for the signal to travel from the sender to the receiver.
- Depends on the distance and transmission medium.

Transmission Delay (12 ms)

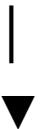
- Time required to place all packet bits onto the communication link.
- Depends on packet size and bandwidth.

Illustration

User Data



Application Layer (4 ms)

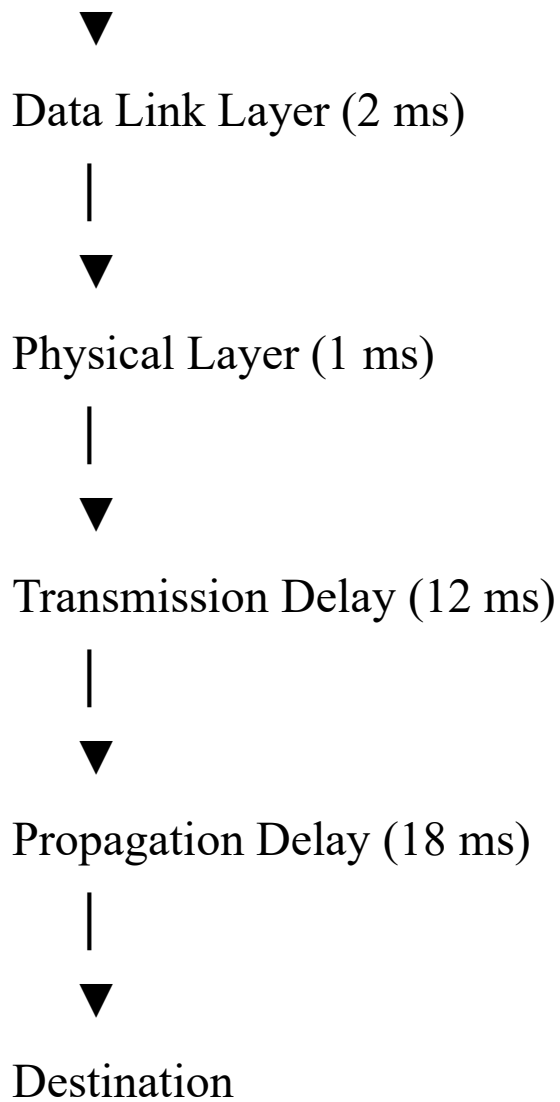


Transport Layer (3 ms)



Network Layer (5 ms)





Total Delay = 45 ms

Real-World Applications

End-to-end delay is important in:

- Video conferencing (Zoom, Google Meet)
- Online gaming
- Voice over IP (VoIP)
- Live video streaming

- Autonomous vehicles
- IoT communication
- Cloud computing

Conclusion

The packet experiences a **processing delay of 15 ms** across the OSI layers. Including the **18 ms propagation delay** and **12 ms transmission delay**, the **total end-to-end delay is 45 ms**. Each OSI layer plays a specific role in ensuring reliable communication, from processing application data to transmitting signals across the physical medium.

Question 9

Problem Statement

A network transfers **80 MB** of useful data using **frames of 1600 bytes**, where each frame contains **80 bytes of protocol overhead** added by different OSI layers.

Calculate:

1. Total number of frames required.
2. Total protocol overhead transmitted.
3. Transmission efficiency as a percentage.
4. Explain how protocol overhead affects network performance.

Solution

Step 1: Given Data

Parameter	Value
Useful Data	80 MB
Frame Size	1600 bytes
Protocol Overhead per Frame	80 bytes

Step 2: Calculate Useful Data per Frame

Each frame has **1600 bytes** in total.

Protocol overhead = **80 bytes**

Therefore,

Formula

$$\text{Useful Data per Frame} = \text{Frame Size} - \text{Overhead}$$

Calculation

$$\begin{aligned} &= 1600 - 80 \\ &= 1520 \text{ bytes} \end{aligned}$$

Answer

$\text{Useful Data per Frame} = 1520 \text{ bytes}$

Step 3: Convert Useful Data into Bytes

We know,

$$1 \text{ MB} = 1024 \times 1024 = 1,048,576 \text{ bytes}$$

Therefore,

$$80 \times 1,048,576 = 83,886,080 \text{ bytes}$$

Answer

$\text{Useful Data} = 83,886,080 \text{ bytes}$

Step 4: Calculate Total Number of Frames

Formula

$$\text{Number of Frames} = \left\lceil \frac{\text{Useful Data}}{\text{Useful Data per Frame}} \right\rceil$$

Calculation

$$= \frac{83,886,080}{1520} = 55,188.21$$

Since a partial frame is not possible,

$\text{Number of Frames} = 55,189$

Step 5: Calculate Total Protocol Overhead

Each frame adds **80 bytes**.

Formula

$$\text{Total Overhead} = 55,189 \times 80$$

Calculation

$$= 4,415,120 \text{ bytes}$$

Convert into MB

$$\frac{4,415,120}{1,048,576} = 4.21 \text{ MB}$$

Answer

$\text{Total Protocol Overhead} = 4,415,120 \text{ bytes } (\approx 4.21 \text{ MB})$

Step 6: Calculate Total Data Transmitted

Useful Data

$$83,886,080 \text{ bytes}$$

•

Protocol Overhead

$$4,415,120 \text{ bytes}$$

=

$$88,301,200 \text{ bytes}$$

Answer

$$\text{Total Data Transmitted} = 88,301,200 \text{ bytes}$$

Step 7: Calculate Transmission Efficiency

Formula

$$\text{Transmission Efficiency} = \frac{\text{Useful Data}}{\text{Total Data Transmitted}} \times 100$$

Calculation

$$\begin{aligned} &= \frac{83,886,080}{88,301,200} \times 100 \\ &= 94.99\% \end{aligned}$$

≈

$$\text{Transmission Efficiency} = 95\%$$

Final Answers

Calculation	Answer
Useful Data	80 MB
Useful Data per Frame	1520 bytes
Number of Frames	55,189
Total Protocol Overhead	4,415,120 bytes (≈4.21 MB)

Calculation

Answer

Total Data Transmitted **88,301,200 bytes**

Transmission Efficiency **95%**

Concept Explanation

What is Protocol Overhead?

Protocol overhead is the **extra information** added to user data by different OSI layers to ensure proper communication.

Examples include:

- Headers
- Trailers
- Error detection bits
- Sequence numbers
- Address information

Although overhead does not contain user data, it is essential for reliable communication.

How Protocol Overhead Affects Network Performance

Advantages

- Enables error detection and correction.
- Supports reliable data delivery.
- Provides addressing and routing information.
- Improves communication security.

- Ensures proper packet sequencing.

Disadvantages

- Increases total data transmitted.
- Consumes additional bandwidth.
- Reduces transmission efficiency.
- Slightly increases transmission time.

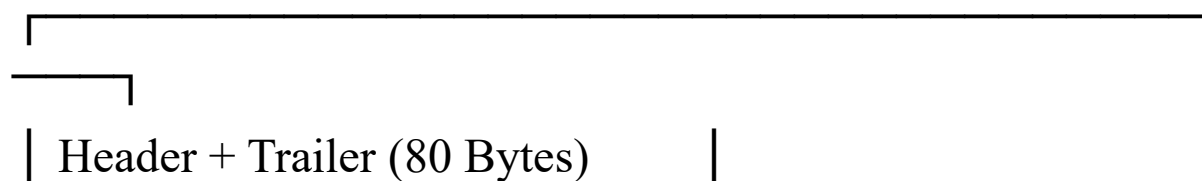
Real-World Applications

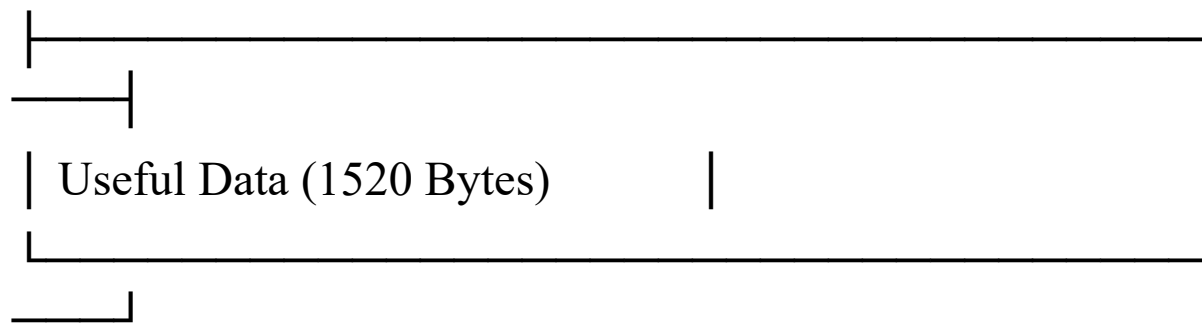
Protocol overhead is important in:

- Internet communication
- Cloud computing
- Video streaming
- Online banking
- Email services
- IoT devices
- 5G communication

Illustration

Frame (1600 Bytes)





55,189 Frames



Useful Data = 80 MB

Overhead = 4.21 MB

Transmission Efficiency = 95%

Conclusion

The network transfers **80 MB** of useful data using **55,189 frames**. Each frame includes **80 bytes of protocol overhead**, resulting in a total overhead of **4,415,120 bytes (approximately 4.21 MB)**. The overall **transmission efficiency is approximately 95%**, meaning most of the transmitted data is useful payload while a small portion is protocol overhead required for reliable communication.

Question 10

Problem Statement

A hospital transfers a patient's **240 MB** medical record through an **IoT-enabled healthcare network**.

- The **Presentation Layer** compresses the data by **30%**.
- The **Transport Layer** divides the compressed data into **1400-byte segments**.
- The **Data Link Layer** adds **60 bytes** of overhead to each frame.

Calculate:

1. The compressed file size.
2. The total number of segments transmitted.
3. The total protocol overhead introduced.
4. The final amount of data transmitted across the Physical Layer.
5. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Solution

Step 1: Given Data

Parameter	Value
Original File Size	240 MB
Compression	30%
Segment Size	1400 bytes
Data Link Layer Overhead	60 bytes/frame

Step 2: Calculate Compressed File Size

The Presentation Layer compresses the file by **30%**.

Formula

$$\text{Compressed Size} = \text{Original Size} \times (1 - \text{Compression})$$

Calculation

$$\begin{aligned} &= 240 \times (1 - 0.30) \\ &= 240 \times 0.70 \\ &= 168 \text{ MB} \end{aligned}$$

Answer

$\text{Compressed File Size} = 168 \text{ MB}$
--

Step 3: Convert Compressed File into Bytes

We know,

$$1 \text{ MB} = 1024 \times 1024 = 1,048,576 \text{ bytes}$$

Therefore,

$$168 \times 1,048,576 = 176,160,768 \text{ bytes}$$

Answer

$\text{Compressed File} = 176,160,768 \text{ bytes}$
--

Step 4: Calculate Number of Segments

Each Transport Layer segment carries **1400 bytes**.

Formula

$$\text{Number of Segments} = \left\lceil \frac{\text{Compressed File Size}}{\text{Segment Size}} \right\rceil$$

Calculation

$$= \frac{176,160,768}{1400} = 125,829.12$$

Since a partial segment is not possible,

$$\boxed{\text{Number of Segments} = 125,830}$$

Step 5: Calculate Total Protocol Overhead

Each frame adds **60 bytes** of overhead.

Formula

$$\text{Total Overhead} = 125,830 \times 60$$

Calculation

$$= 7,549,800 \text{ bytes}$$

Convert to MB:

$$\frac{7,549,800}{1,048,576} = 7.20 \text{ MB}$$

Answer

Total Protocol Overhead = 7,549,800 bytes (≈ 7.20 MB)
--

Step 6: Calculate Final Data Transmitted

Compressed Data

176,160,768 bytes

•

Protocol Overhead

7,549,800 bytes

=

183,710,568 bytes

Convert into MB

$$\frac{183,710,568}{1,048,576} \approx 175.20 \text{ MB}$$

Answer

Final Data Transmitted = 183,710,568 bytes (≈ 175.20 MB)

Final Answers

Calculation	Answer
Original File Size	240 MB
Compressed File Size	168 MB
Number of Segments	125,830
Protocol Overhead	7,549,800 bytes (≈7.20 MB)
Final Data Transmitted	183,710,568 bytes (≈175.20 MB)

Concept Explanation

How Multiple OSI Layers Work Together

1. Presentation Layer (Layer 6)

Functions:

- Compresses data to reduce file size.
- Encrypts sensitive medical records.
- Converts data into a standard format.

Benefit:

- Faster transmission.
- Reduced bandwidth usage.
- Improved data security.

2. Transport Layer (Layer 4)

Functions:

- Divides data into **1400-byte segments**.
- Provides reliable end-to-end communication.
- Performs error detection, retransmission, and flow control.

Benefit:

- Ensures complete and error-free delivery.

3. Data Link Layer (Layer 2)

Functions:

- Encapsulates each segment into a frame.
- Adds **60-byte header and trailer**.
- Detects transmission errors using CRC.

Benefit:

- Reliable communication between directly connected devices.

4. Physical Layer (Layer 1)

Functions:

- Converts frames into electrical, optical, or wireless signals.
- Transmits data through cables or wireless networks.

Benefit:

- Enables actual data transmission over the communication medium.

How the Layers Work Together

Patient Medical Record (240 MB)



Presentation Layer

Compression (30%)



168 MB



Transport Layer

Segments (1400 Bytes)



125,830 Segments



Data Link Layer

Adds 60 Bytes Overhead



Physical Layer

Transmits 175.20 MB



Hospital Server

Real-World Applications

This communication process is used in:

- Smart Hospitals
- IoT Healthcare Systems
- Electronic Health Records (EHR)
- Telemedicine
- Cloud Healthcare Platforms
- Medical Imaging Systems
- Remote Patient Monitoring

Conclusion

The **240 MB** medical record is compressed to **168 MB** by the **Presentation Layer**. The **Transport Layer** divides the compressed data into **125,830 segments**, while the **Data Link Layer** adds **7,549,800 bytes (approximately 7.20 MB)** of protocol overhead. As a result, the **Physical Layer transmits**.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers